

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Quantities, Units & Modelling

*Mechanics begins by naming the quantity, unit, and model
before doing any algebra.*

QUANTITIES, UNITS & MODELLING · M1 · 01

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The Map

This strategy booklet was mined from the local Mechanics 1 topic problem/answer PDFs, Dr Eslam's source notes, and the WME01 website answer bank. Mechanics questions become reliable when the diagram, sign convention, and equation family are chosen first.

01 UNIT: Unit Discipline

TRIGGER *units, modelling, dimensions*

02 MODEL: Modelling Words

TRIGGER *particle, light string, smooth plane*

03 WEIGH: Mass And Weight

TRIGGER *mass, weight, g*

04 ROUND: Accuracy Control

TRIGGER *show that, 3 significant figures*

BANK EVIDENCE Local pages: 29 problem, 46 answer, 15 notes. Website primary entries: 0. Website linked entries: 42.

01 UNIT: Unit Discipline

TRIGGER *units, modelling, dimensions*

BECOMES Convert and carry units before substituting.

FIRST LINE TO WRITE

$$\text{quantity} = \text{number} \times \text{unit}$$

SIMPLEST STRATEGY

1. Underline the quantity.
2. Name the unit.
3. Insert conversions.
4. Test the final unit.

WORKED MODEL

$$\text{I } 72 \text{ km h}^{-1} = 20 \text{ m s}^{-1}.$$

HARD VARIANTS

1. Convert before using SUVAT or $F = ma$.
2. For compound units, convert every component of the unit.
3. Check if the question fixes $g = 9.8$ or allows exact g .

— BOTTOM LINE

Units are part of the answer, not decoration.

COMMON TRAP Mixing metres with kilometres or hours with seconds.

02 MODEL: Modelling Words

TRIGGER *particle, light string, smooth plane*

BECOMES Translate each modelling word into a mathematical permission.

FIRST LINE TO WRITE

| smooth $\Rightarrow F = 0$

SIMPLEST STRATEGY

1. Mark the modelling word.
2. Output its meaning.
3. Draw what remains.
4. Exclude ignored effects.
5. Link to the equation.

WORKED MODEL

| Light string means the tension is the same throughout that string.

HARD VARIANTS

1. Smooth means no friction, not no normal reaction.
2. Light means no weight for that string or rod only.
3. Inextensible means connected particles have the same acceleration magnitude.

— BOTTOM LINE

A modelling word is an instruction for the diagram.

03 WEIGH: Mass And Weight

TRIGGER *mass, weight, g*

BECOMES Convert mass into weight only when a force is needed.

FIRST LINE TO WRITE

$$W = mg$$

SIMPLEST STRATEGY

1. Write $W = mg$.
2. Enter mass in kg.
3. Insert g .
4. Give force in newtons.
5. Hold exact g if requested.

WORKED MODEL

■ A 5 kg particle has weight 5g N, or 49 N if $g = 9.8$.

HARD VARIANTS

1. Use mg in moments and dynamics, not just m .
2. If answers use g , do not replace it unless asked.
3. For grams, convert to kg before multiplying by g .

— BOTTOM LINE

Mass is measured in kg; weight is a force.

04 ROUND: Accuracy Control

TRIGGER *show that, 3 significant figures*

BECOMES Keep exact working, round only the final statement.

FIRST LINE TO WRITE

| exact first, rounded last

SIMPLEST STRATEGY

1. Retain exact fractions or radicals.
2. Only round at the end.
3. Use the requested accuracy.
4. Name units.
5. Do a reasonableness check.

WORKED MODEL

| $\sqrt{52} = 7.211\dots$, so 7.21 to 3 s.f.

HARD VARIANTS

1. For show-that answers, keep enough figures to support the printed value.
2. Bearings need three digits.
3. Do not mix exact g and decimal g in the same final line.

— BOTTOM LINE

Rounding early moves the answer.

COMMON TRAP Rounding inside the equation and losing the mark.

Quick Recognition Table

WHEN YOU SEE	FIRST LINE TO WRITE
units, modelling, dimensions	quantity = number $\times$ unit
particle, light string, smooth plane	smooth $\Rightarrow F = 0$
mass, weight, g	$W = mg$
show that, 3 significant figures	exact first, rounded last

Hard Variant Checklist

STRATEGY	CHECK BEFORE FINAL ANSWER
UNIT: Unit Discipline	Convert before using SUVAT or $F = ma$. For compound units, convert every component of the unit. Check if the question fixes $g = 9.8$ or allows exact g .
MODEL: Modelling Words	Smooth means no friction, not no normal reaction. Light means no weight for that string or rod only. Inextensible means connected particles have the same acceleration magnitude.
WEIGH: Mass And Weight	Use mg in moments and dynamics, not just m . If answers use g , do not replace it unless asked. For grams, convert to kg before multiplying by g .
ROUND: Accuracy Control	For show-that answers, keep enough figures to support the printed value. Bearings need three digits. Do not mix exact g and decimal g in the same final line.